

- 7 (a) When 2 or more waves of the same kind meet at a point, they superpose to produce a resultant wave whereby there is a change in the total displacement. B1

Comments:

This part is also badly done by most student. Most of them did not use the key phrases such as “of the same kind” or “superpose” in defining the term.

$$(b) \lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{1.0 \times 10^{10}} = 0.030 \text{ m} \quad \text{M1}$$

$$\text{Aerial length} = \frac{\lambda}{2} = \frac{0.030}{2} = 0.015 \text{ m} \quad \text{A1}$$

Comments:

This part is generally well done by most students. Only a few of them have problems with the conversion of units from GHz to Hz.

- (c) (i) 1. The path difference between the signal changes along OP. B1

When the path difference is equal = $n\lambda$, where n is an integer, .
(OR when signals meet in phase at points along OP)

constructive interference takes place, and the resultant amplitude is the sum of the individual amplitudes. B1

When the path difference is equal = $(n-1/2)\lambda$, where n is an integer, . (OR when signals meet in anti-phase at points along OP)

destructive interference takes place, and the resultant amplitude is the difference of the 2 amplitudes. B1

Comments:

Most of the students who attempted this were able to quote the conditions for constructive and destructive interference.

$$7(c)(i) \quad 2. \quad \Delta x = \lambda \frac{D}{a} = 0.030 \times \frac{4.0}{0.20} = 0.60 \text{ m} \quad \text{C1}$$

$$\text{Distance} = \frac{\Delta x}{2} = 0.30 \text{ m} \quad \text{A1}$$

Comment:

Students must realise that the distance between max and min is $\frac{1}{2}$ the fringe separation.

- (ii) As detector is moved from O to Q, the amplitude of signal from both A and B decreases since amplitude varies inversely with distance B1

Because energy emitted by the transmitters spread out (so less is collected by the receiver the further away it is) B1

OR

Since intensity varies inversely with the square of separation, as the detector is moved from O to Q, the intensity from each transmitter A and B decreases.

As intensity is proportional to the square of amplitude, since intensity at Q decreases, the amplitude from both A and B decrease and hence the resultant amplitude at Q decreases.

Comments:

This part of the question is not well explained. Many students used symbols r and A with explaining what they represent.

- (iii) 1. When the path difference AO and BO is $\lambda/2$, a minimum occurs due to destructive interference between the signals from A and B. When the path difference AO and BO is λ , a maximum occurs due to constructive interference between the signals from A and B.

B1

$$\begin{aligned} 2. \text{ distance} &= \frac{\lambda}{2} && \text{C1} \\ &= \frac{0.030}{2} = 0.015 \text{ m} && \text{A1} \end{aligned}$$

Comments:

Again this part is not done well due to proper concept.

- (d) (i) Intensity increases by a factor of 4 B1
As intensity $\propto$ (amplitude)² B1

Comments:

$$\frac{I'}{I} = \left(\frac{2A + 2A}{A + A} \right)^2 = 4$$

- (ii) Intensity becomes 0 instead of maximum B1
Path difference AO – BO = 0, and so destructive interference occurs at O,
and a minima will be at O. B1

Comments:

Waves arriving at O travelled through the same path length.

Since they were anti-phase to each other at A and B, they arrive anti-phase to each other at O.

- (iii) Intensity falls to zero B1
Emitted signals is vertically polarized B1
Receiver in position is only able to detect horizontally polarized signal. B1

Comments:

There is no change in the plane of polarization from the 2 sources.

When the detector is rotated 90° , the plane of polarization of the resultant wave is 90° to the transmission axis of the detector. Intensity detected becomes $I \cos^2 90$